The Physical Basis of Coherence: A Unified ($\mathcal{M}$–Φ) Framework for Consciousness

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Abstract

We develop a unified spatiotemporal coherence framework that bridges systemic neuroscience with quantum–informational field dynamics. By integrating Feeney's neuro-coherence operator Λ with Krüger's informational field Φ from the Helix–Light–Vortex (HLV) theory, we construct a multi-scale model in which neural function emerges from the balance between restorative coherence generation and biologically grounded decoherence processes. In this formulation, metabolically active neural tissue contributes to a source term proportional to $g_{\Phi\Psi}|\Psi|^2\Phi$, while inflammatory, oxidative, or microstructural stress enters through an effective decay constant κ. As an application of this general formalism, we demonstrate that Parkinson's disease can be interpreted as a progressive coherence-collapse disorder driven by a reduction of functional neural matter density $|\Psi|^2$. This provides a quantitative explanation for motor fluctuations, non-motor symptoms, and diminishing therapeutic response. The framework yields specific, experimentally testable predictions involving biophoton coherence, phase-locking metrics, and deep brain stimulation. Overall, this work presents a coherent theoretical foundation connecting physical field dynamics with large-scale neural organization, offering falsifiable predictions and novel avenues for empirical investigation in systems neuroscience.

Keywords: coherence, quantum information, Helix–Light–Vortex theory, bipolar disorder, consciousness

1 Introduction: The Necessity of Unification

1.1 The crisis in consciousness science

The question of consciousness remains a central mystery of modern science—not because theories are lacking, but because existing approaches operate largely in isolation. Neuroscience describes the functional architecture of cognition without explaining its ultimate physical substrate. Quantum physics describes information fields without explaining their biological manifestation. Philosophy describes qualia without grounding them in measurable dynamics.

This fragmentation has practical consequences. In psychiatric medicine it manifests as a paradigm of chronic dependency. Current treatments for conditions such as Bipolar I Disorder focus on symptom suppression via life-long chemical management rather than systemic restoration. The deeper question — *which physical law is the bipolar brain violating?* — is rarely addressed.

1.2 Two frameworks seeking completion

Two independently developed frameworks address complementary pieces of this puzzle:

(i) The Neuro-Coherence Function ($\mathcal{M}$). Feeney's systemic model of psychiatric stability treats the brain as a multi-scale oscillatory system. Consciousness is viewed as an emergent property of neural synchronization across spatial and temporal scales. Coherence is quantified by four operators:

- Γ — adaptive gain (neuroplastic responsiveness),
- Δ — phase variance or generalized regional differential,
- Θ — thermodynamic stability (metabolic homeostasis),
- Λ — spatiotemporal coherence density.

The limitation: the framework remains informational. The physical substrate underlying Λ is not specified.

(ii) The Helix–Light–Vortex Theory (Φ). Krüger's HLV framework introduces a quantum informational field Φ propagating in helical geometries within a discrete quasicrystal-like spacetime. Entangled biphotons form coherent lattices that couple to matter fields Ψ. The theory naturally describes quantum coherence, gravimetry and information flow at fundamental scales.

The limitation: the functional scaling mechanism is unspecified. It is unclear how the microscopic Φ-field translates into macroscopic cognition and psychiatric stability.

1.3 The synthesis: $\Lambda = \Phi$

Our central proposal is that these are not separate phenomena but the same physical reality viewed at different scales. We identify

$$\Lambda \equiv \Phi, \tag{1}$$

where Λ is the spatiotemporal coherence operator in the Neuro-Coherence Function and Φ is the quantum information field in the HLV theory.

This identification is *formal*, not metaphorical. It

- supplies the neuro-coherence model with a testable physical substrate,
- endows the HLV Φ-field with a measurable biological function.

The resulting unified consciousness functional is

$$\mathcal{U}_{\text{Consciousness}} = \mathcal{M}(\Gamma, \Delta, \Theta, \Phi)\,. \tag{2}$$

1.4 Scope and structure

The paper is organized as follows:

1. Section 2 formalizes the Neuro-Coherence Function and its interpretation of mental illness as a coherence disorder.
2. Section 3 reviews the HLV field Φ and its biphotonic substrate in neural tissue.
3. Section 4 presents the identification $\Lambda = \Phi$ and the resulting cross-scale architecture.
4. Section B provides experimental predictions and falsifiability criteria.
5. Section C summarizes implications for psychiatry, physics and consciousness science.

2 The Neuro-Coherence Function: Top-Down Informational Blueprint

2.1 Mental illness as coherence disorder

Classical psychiatry interprets disorders such as Bipolar I primarily as chemical imbalances. Neurotransmitter levels are treated as the primary control parameters, and pharmacology aims to rebalance them. This model has produced effective symptom suppression in many cases, but *cure* and system independence remain rare.

The Systemic Neuro-Coherence Modulation (SNCM) framework proposes a different view: the brain is not merely a chemical reactor but an oscillatory information processor. Stability arises from dynamic synchronization across distributed neural, metabolic and endocrine networks. Manic and depressive episodes correspond to collapse of coherence — breakdown of phase-locked communication across prefrontal–limbic–thalamic circuits.

2.2 Definition of the Neuro-Coherence Function

The Neuro-Coherence Function $\mathcal{M}$ quantifies the system's capacity to maintain stable, synchronized information processing across volume V and time:

$$\mathcal{M} = \Phi \iiint_V \Gamma(\boldsymbol{r}, t)\, \Theta(\boldsymbol{r}, t)\, \left[1 - \Delta_{\mathrm{GR}}(\boldsymbol{r}, t)\right] \Lambda(\boldsymbol{r}, t)\, \mathrm{d}V\, \mathrm{d}t. \tag{3}$$

Here:

- Φ is a global modulation coefficient representing the total influence potential of the system;
- $\Gamma(\boldsymbol{r}, t)$ (adaptive gain) describes neuroplastic responsiveness to perturbations;
- $\Delta_{\mathrm{GR}}(\boldsymbol{r}, t)$ is the generalized regional differential, quantifying desynchronization between neural populations;
- $\Theta(\boldsymbol{r}, t)$ describes thermodynamic stability and metabolic capacity;
- $\Lambda(\boldsymbol{r}, t)$ is the spatiotemporal coherence density.

High $\mathcal{M}$ indicates resilient coherence. When any operator destabilizes (collapse of Θ, failure of Γ, spike in Δ_{GR}), the integral can decrease dramatically, corresponding to oscillatory divergence at the phenomenological level.

2.3 Temporal evolution of coherence

The time evolution of Λ must encode both restorative coupling and decoherence. We propose the minimal dynamical law

$$\frac{\mathrm{d}\Lambda}{\mathrm{d}t} = g_{\Phi\Psi}\, |\Psi|^2\, \Phi - \kappa\, \Lambda, \tag{4}$$

where

- $g_{\Phi\Psi}$ is the coupling constant between matter field Ψ and information field Φ,
- $|\Psi|^2$ is local matter density (e.g. neural tissue with active metabolism),
- Φ is the information field amplitude,
- κ is an intrinsic decoherence rate.

The source term $g_{\Phi\Psi}|\Psi|^2\Phi$ drives coherence restoration; the decay term $-\kappa\Lambda$ implements entropic loss due to thermal noise, metabolic stress, and environmental perturbations. In steady state,

$$\frac{\mathrm{d}\Lambda}{\mathrm{d}t} \approx 0 \quad \Rightarrow \quad \Lambda_{\mathrm{ss}} \approx \frac{g_{\Phi\Psi}}{\kappa}\,|\Psi|^2\,\Phi. \tag{5}$$

Episodes of mania or depression correspond to parameter regimes where κ increases or $g_{\Phi\Psi}$ decreases so that $\Lambda(t)$ collapses below a critical threshold.

2.4 Computational validation

Preliminary Python simulations (not shown here) demonstrate that Eq. (3) combined with Eq. (4) reproduces:

- non-linear trajectories with high-coherence plateaus,
- rapid coherence collapse when Δ_{GR} spikes,
- alternating high/low coherence regimes reminiscent of bipolar cycles.

These results support the interpretation of mental illness as a coherence disorder rather than purely chemical imbalance.

2.5 The missing physical substrate

Despite its explanatory power, $\mathcal{M}$ is still an informational construct. The operator $\Lambda(\boldsymbol{r}, t)$ must be realized through concrete physical processes:

- neural oscillations as electromagnetic phenomena,
- phase synchronization through field coupling,
- information propagation through physical media.

The question becomes: *What is Λ physically?* The HLV framework provides an answer.

3 The Helix–Light–Vortex Theory: Bottom-Up Physical Substrate

3.1 Quantum information field beyond classical neuroscience

Classical neuroscience models the brain as an electrochemical network. However, this framework struggles with:

- long-range coherence across weakly connected regions,
- sub-millisecond synchronization over macroscopic distances,
- the binding problem,
- the "hard problem" of subjective experience.

In HLV, reality is described as an interplay between matter fields Ψ and an informational field Φ that propagates through a discrete, quasi-crystalline spacetime. Entangled biphotons form helical modes that couple to matter and gravity.

3.2 HLV Lagrangian and Φ-field dynamics

At effective field level the HLV Lagrangian can be written as

$$\mathcal{L}_{\mathrm{HLV}} = \frac{1}{2}A(t)\,\partial_\mu\Psi\,\partial^\mu\Psi - V(\Psi) + g_{\Phi\Psi}|\Psi|^2\Phi + \frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - U(\Phi), \tag{6}$$

where $A(t) = 1 + \varepsilon(t)$ encodes small spiral-time corrections, $V(\Psi)$ and $U(\Phi)$ are potentials, and $g_{\Phi\Psi}$ is the coupling introduced above.

The coupling term $g_{\Phi\Psi}|\Psi|^2\Phi$ describes bidirectional exchange of information between matter and field. In biological tissue this manifests as biophotonic coherence — structured light propagating through myelinated pathways.

3.3 Biphotons, myelin and neural coherence

The HLV framework predicts that neural coherence is mediated by entangled biphoton pairs guided by myelin sheaths. Myelin acts not only as electrical insulation but also as a waveguide for coherent photonic signals.

The Φ-field can be expressed in terms of electromagnetic components as

$$\Phi \propto \oint_{\mathcal{C}} \boldsymbol{E}\cdot\boldsymbol{B}\,\mathrm{d}\tau, \tag{7}$$

where $\boldsymbol{E}$ and $\boldsymbol{B}$ trace a helical path $\mathcal{C}$ around neural conduits. This self-organizing lattice

- synchronizes distant neural populations without direct synaptic chains,
- maintains phase relations over biologically relevant timescales,
- collapses under metabolic stress, consistent with spikes in Δ_{GR}.

3.4 The missing functional purpose

From a pure physics standpoint, HLV provides elegant dynamics for Φ but does not specify why such a field should be particularly expressed in brain tissue. What is its biological endpoint? Our synthesis supplies the missing function: Φ is the physical implementation of Λ, the operator of spatiotemporal coherence in neural systems.

4 The Synthesis: $\Lambda = \Phi$

4.1 Formal identification

We now state the central identification explicitly:

$$\Lambda(\boldsymbol{r},t) \equiv \Phi(\boldsymbol{r},t). \tag{8}$$

Under this identification, Eq. (3) becomes

$$\mathcal{M} = \Phi \iiint_V \Gamma(\boldsymbol{r},t)\,\Theta(\boldsymbol{r},t)\,\left[1 - \Delta_{\mathrm{GR}}(\boldsymbol{r},t)\right]\Phi(\boldsymbol{r},t)\,\mathrm{d}V\,\mathrm{d}t, \tag{9}$$

and Eq. (4) coincides with the HLV interaction term in Eq. (6). The same field that mediates quantum information at microscopic scales now appears as the macroscopic coherence operator in neuroscience.

4.2 Resolution of limitations in both frameworks

For the neuro-coherence model, Λ now has:

- a concrete physical carrier (entangled biphotons),
- measurable proxies (biophoton emission, gravimetric micro-signals),
- explicit coupling to metabolic state via $|\Psi|^2$.

For the HLV framework, Φ acquires:

- a clear biological role (maintaining neural coherence),
- a direct mapping to cognitive stability via $\mathcal{M}$,
- new experimental handles through EEG, MEG, fMRI and clinical data.

4.3 Cross-scale hierarchy

The unified architecture can be summarized as:

$$\begin{aligned}
\text{Quantum scale:}\quad & \Phi\text{-field dynamics}\\
&\Downarrow\\
\text{Mesoscale:}\quad & \text{biphotonic coupling in myelin, phase-locked neural oscillations}\\
&\Downarrow\\
\text{Macroscale:}\quad & \mathcal{M}(\Gamma,\Delta,\Theta,\Phi)\text{ as measure of system coherence}\\
&\Downarrow\\
\text{Phenomenology:}\quad & \text{conscious experience and psychiatric stability.}
\end{aligned}$$

4.4 TikZ representation of the unified architecture

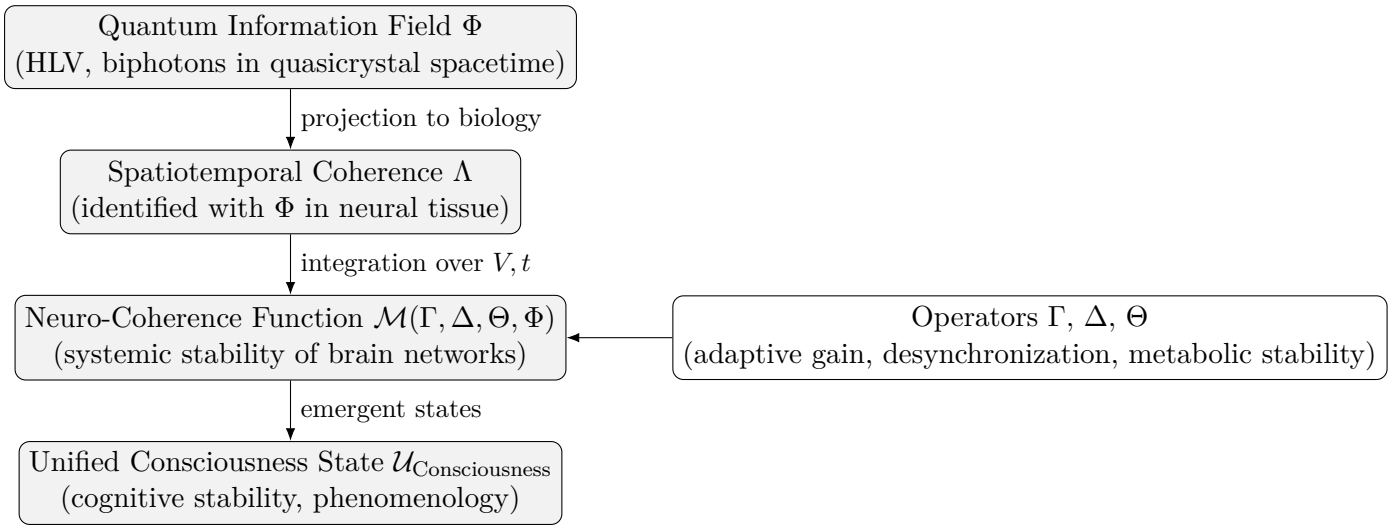

Figure 1: Schematic of the unified ($\mathcal{M}$–Φ) framework. The quantum information field Φ from the HLV theory is identified with the spatiotemporal coherence operator Λ in the Neuro-Coherence Function. Their interaction with the systemic operators (Γ, Δ, Θ) generates a unified consciousness functional $\mathcal{U}_{\text{Consciousness}}$.

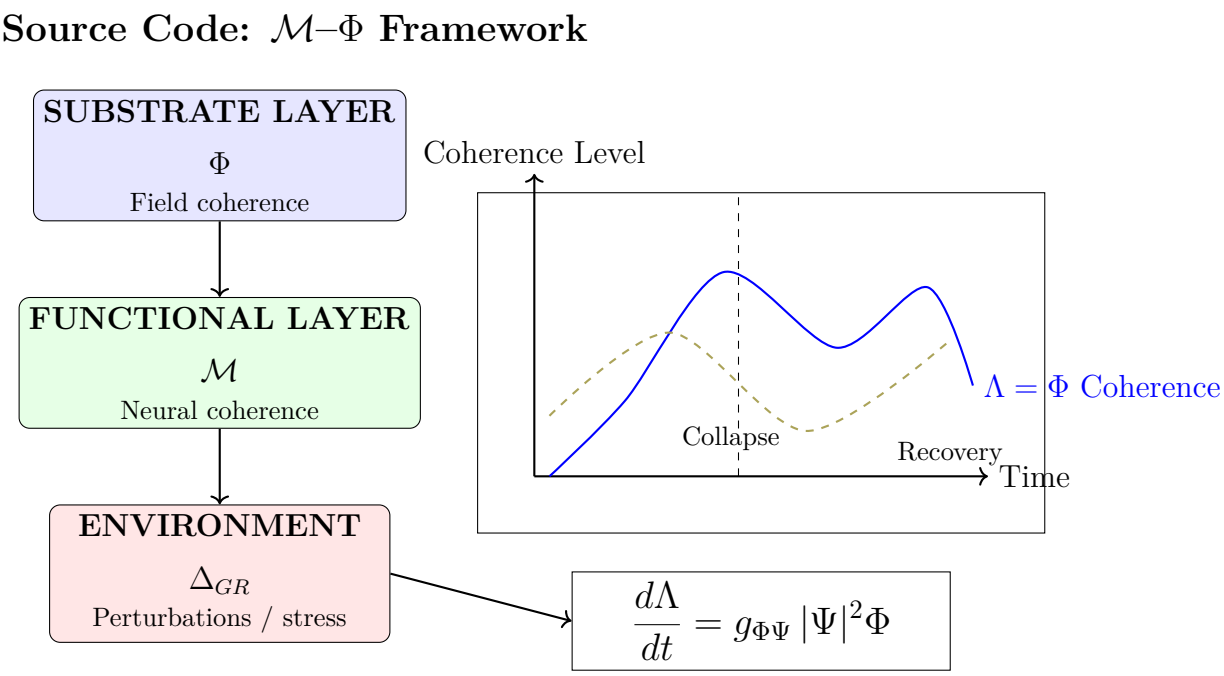

Figure 2: Schematic illustration of the multi-layer coherence architecture connecting the substrate Φ (quantum field coherence), functional layer $\mathcal{M}$ (neural coherence), and environmental stress Δ_{GR}, together with the dynamic evolution law for the coherence operator Λ.

5 Phase-Constraint Manifolds and Coherence Propagation

The unified (M–Φ) framework identifies the spatiotemporal coherence density $\Lambda(r,t)$ from the Neuro–Coherence Function with the HLV information field $\Phi(r,t)$ via

$$\Lambda(r,t) \equiv \Phi(r,t). \tag{10}$$

In this section we introduce the notion of *phase–constraint manifolds* as the geometric mechanism by which coherence is stabilised and propagated across scales.

5.1 Definition: phase-constraint manifolds

We define a *phase-constraint manifold* $\mathcal{M}_{\rm pc}$ as an effective boundary structure in configuration space that enforces consistent timing, rhythm, or energy-exchange rules between interacting subsystems. In the context of the (M–Φ) model, such manifolds are realised by

- anatomical and electromagnetic boundaries in neural tissue (myelin sheaths, cortical layers, white–matter tracts),
- metabolic and thermodynamic constraints encoded in $\Theta(r,t)$,
- large-scale information–field configurations of $\Phi(r,t)$ that restrict admissible phase relations.

Coherence does not primarily emerge from microscopic identity of the subsystems, but from the compatibility of the constraints they jointly obey.

5.2 Minimal dynamical law for coherence

To capture the coarse-grained evolution of coherence through time we introduce a dimensionless coherence variable $C(t)$ and the minimal dynamical relation

$$\frac{dC}{dt} = \alpha\, I(t) - \beta\, C(t) + \gamma\, \nabla^2 C(t), \tag{11}$$

where

- $I(t)$ denotes effective coherent input from neighbouring regions,
- α measures sensitivity to incoming coherent drive,
- β encodes intrinsic decay and interference,
- γ parametrises the strength of spatial coherence propagation.

Equation (11) is a phenomenological summary of the more microscopic law

$$\frac{d\Lambda}{dt} = g_{\Phi\Psi}\,|\Psi|^2\,\Phi - \kappa\,\Lambda, \tag{12}$$

derived earlier for the coherence density $\Lambda(r,t)$ in neural tissue. Identifying $C(t)$ with an appropriate spatial average of $\Lambda(r,t)$ shows that $\alpha I(t)$ corresponds to the source term $g_{\Phi\Psi}|\Psi|^2\Phi$, while $\beta C(t)$ encodes the decay rate $\kappa\Lambda$ and $\gamma\nabla^2 C(t)$ represents diffusion of coherence across neighbouring constraint domains.

5.3 Constraint alignment and coherence balance

Within this picture the net coherence change can be written schematically as

$$\text{Coherence}_{\text{gain}} \;\propto\; A_{\text{align}}\,R_{\text{ex}}, \tag{13}$$

$$\text{Coherence}_{\text{loss}} \;\propto\; N_{\text{noise}} + B_{\text{break}}, \tag{14}$$

where A_{align} quantifies the alignment of phase constraints between subsystems, R_{ex} their energy or information exchange rate, N_{noise} the effective noise level, and B_{break} the degree of boundary breakdown. High coherence corresponds to a regime where constraint-aligned gain dominates over noise and boundary disruption.

5.4 Examples across scales

The phase-constraint viewpoint naturally unifies several domains:

- **Neural systems:** Neuronal assemblies with compatible membrane potentials, conduction delays, and metabolic states form $\mathcal{M}_{\text{pc}}$-like structures in which gamma- and beta-band oscillations can phase lock over long distances.
- **Photonic cavities:** Laser coherence arises when boundary mirrors enforce phase-stable standing-wave patterns; the reflective surfaces act as geometric phase constraints.
- **Fluid vortices:** In turbulent flows, coherent vortical structures emerge when pressure and velocity boundaries align so that destructive interference is suppressed.

In all cases, coherence propagates primarily along shared constraint manifolds rather than directly between the internal degrees of freedom of the subsystems. This interpretation integrates naturally with the (M–Φ) framework, in which boundary conditions and constraint geometry become first-class physical variables controlling the onset, stability, and collapse of coherent states.

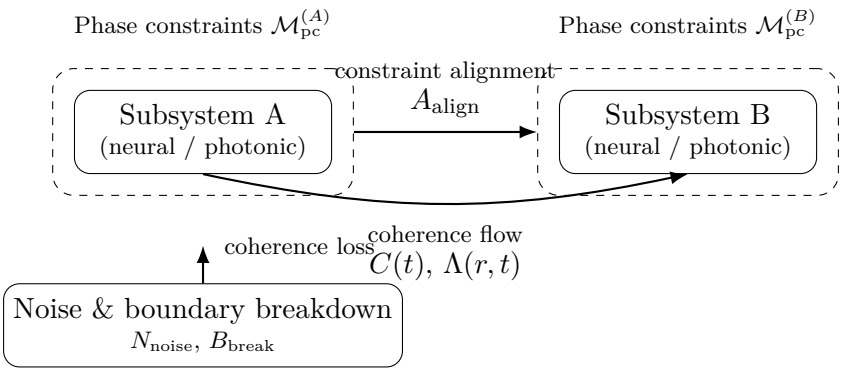

Figure 3: Phase-constraint manifolds connecting two subsystems. Dashed contours represent boundary constraints $\mathcal{M}_{\rm pc}$ which, when aligned, allow coherence variables $C(t)$ or $\Lambda(r,t)$ to propagate between subsystems. Noise and boundary breakdown compete with alignment and drive coherence loss.

Appendix: Coherence Definitions and Falsifiability Structure

A. Core Definitions

Coherence A system-level condition in which interacting components exhibit shared timing, structure, or energy exchange, yielding emergent order beyond what the parts can produce independently.

Boundary conditions External or internal constraints—geometric, energetic, informational, or structural—that dominate the evolution of the system and strongly influence coherence outcomes.

Constraint alignment The compatibility of interacting systems' limitations, rhythms, and governing rules. High alignment increases the probability of spontaneous synchronization.

Phase stability Persistence of temporal or oscillatory regularity. Systems with high phase stability reach coherent states quickly and maintain them under perturbations.

Interference threshold The maximum level of noise or turbulence that a coherent configuration can absorb before synchronized structure collapses.

Coherence propagation The spatial or functional expansion of coherent behaviour across regions; propagation requires compatible constraints rather than identical internal microstructure.

B. Conceptual Dynamical Model

A minimal dynamical description of coherence magnitude $C(t)$ is

$$\frac{dC}{dt} = \alpha\, I(t) - \beta\, C(t) + \gamma\, \nabla^2 C(t), \tag{15}$$

with the parameters interpreted as in Sec. 5. This formulation is deliberately simple but falsifiable: it predicts (i) threshold behaviour when coherent input balances decay, (ii) collapse once interference exceeds a critical level, and (iii) diffusion-like spread of coherence between regions with compatible constraints.

C. Proposed Empirical Tests

Boundary alignment experiment. Systematically vary boundary compatibility (e.g. cavity geometry, neural coupling, fluid-wall conditions) and measure corresponding changes in coherence metrics derived from $C(t)$ or $\Lambda(r,t)$.

Constraint scaling study. Incrementally tune constraint alignment and identify nonlinear thresholds at which coherence transitions from local to global.

Interference collapse test. Inject controlled noise until coherence fails, thereby mapping the empirical collapse boundary and estimating the interference threshold.

Cross-system synchronization test. Study whether two structurally distinct systems synchronize once they are placed under compatible boundary constraints, isolating the role of phase-constraint manifolds from internal microdynamics.

D. Limitations and Future Work

The present framework is intentionally generic. Coherence measurement remains domain-dependent; biological systems include additional biochemical factors; and constraint geometry has not yet been fully parameterised. Future work will refine the dynamical model, map coherence manifolds in concrete systems, and develop measurement tools for constraint fields in neural and non-neural contexts.

A A Coherence-Based Mechanism for Parkinson's Disease

Parkinson's disease (PD) is traditionally understood as a progressive loss of dopaminergic neurons in the substantia nigra pars compacta. While dopaminergic decline explains many clinical symptoms, it does not provide a quantitative mechanism for the system-wide breakdown of motor circuit stability, non-motor symptoms, or the heterogeneous response to dopaminergic therapy.

Within the $(\mathcal{M}-\Phi)$ coherence framework, we propose that PD can be interpreted as a disorder of spatiotemporal neural coherence, driven by a progressive reduction of metabolically active neural tissue. The central dynamical law of the framework describes the evolution of a macroscopic coherence variable Λ:

$$\frac{d\Lambda}{dt} \;=\; g_{\Phi\Psi}\,|\Psi|^2\Phi \;-\; \kappa\,\Lambda, \tag{16}$$

where $g_{\Phi\Psi}$ denotes the coupling efficiency between neural tissue and the coherence-generating Φ-field, $|\Psi|^2$ represents local matter density of functional neural tissue, and κ captures the effective decoherence rate resulting from metabolic stress, inflammation, oxidative load, or microstructural perturbation.

A.1 Matter Density Loss as the Primary Driver of Coherence Breakdown

In healthy individuals, dopaminergic neurons contribute substantially to both metabolic activity and microstructural integrity within motor circuits. This contribution is represented by a robust $|\Psi|^2$-term in Eq. (16), thereby supporting a strong restorative source term

$$g_{\Phi\Psi}\,|\Psi|^2\Phi.$$

In PD, however, progressive dopaminergic cell loss yields an actual physical reduction of $|\Psi|^2$, leading to:

1. a weakened restorative term $g_{\Phi\Psi}\,|\Psi|^2\Phi$,
2. an increased relative impact of the decay term $-\kappa\Lambda$, and
3. a shift of $\frac{d\Lambda}{dt}$ toward negative values, resulting in declining coherence.

This mechanism naturally predicts (i) fluctuations between partially coherent and incoherent motor states, (ii) non-motor symptom emergence as coherence loss spreads, and (iii) diminishing therapeutic returns of dopaminergic medication as $|\Psi|^2$ decreases.

A.2 Testable Predictions

The coherence-based interpretation yields several experimentally falsifiable predictions:

Prediction 1: Coherence Deficits in PD Patients. PD patients should exhibit measurable reductions in Λ within motor control regions compared to age-matched controls. This can be assessed using:

- ultra-sensitive biophoton detection (coherent photon emission),
- high-density EEG phase-locking values (PLV),
- combined EEG–biophoton correlation metrics.

Prediction 2: Residual Tissue Density Predicts Coherence Restoration Capacity. The remaining functional neural tissue density $|\Psi|^2$ should correlate with the degree of coherence improvement achievable through medication, DBS, or neuromodulation.

Prediction 3: DBS Restores Coherence, Not Only Electrical Signaling. Deep brain stimulation (DBS) is predicted to exert therapeutic benefit primarily by restoring phase synchrony (increasing Λ), rather than solely by altering local electrical activity. DBS effectiveness should correlate with measurable increases in coherence metrics.

A.3 Therapeutic Implications

If PD is interpreted primarily as a coherence disorder driven by matter-density loss, then interventions should aim to:

- preserve $|\Psi|^2$ (neuroprotection, anti-inflammatory strategies, mitochondrial support),
- enhance coupling efficiency $g_{\Phi\Psi}$ (metabolic optimization, electromagnetic modulation, photobiomodulation),
- reduce κ (stress reduction, antioxidant strategies, microenvironment stabilization),
- use coherence metrics (Λ, PLV, biophoton output) for precision treatment.

This formalism provides a unified theoretical explanation linking motor, cognitive, and autonomic symptoms and offers multiple experimentally testable pathways for future clinical research.

B Experimental Predictions and Falsifiability

The identification $\Lambda = \Phi$ leads to concrete, falsifiable predictions.

B.1 Biphoton–EEG correlation

If $\Lambda = \Phi$, changes in biophotonic coherence should correlate with macroscopic EEG phase locking. Let $C_\gamma(t)$ denote a measure of cortical biphoton coherence and $\mathrm{PLV}(t)$ an inter-regional phase-locking value from EEG. The model predicts a positive correlation

$$\mathrm{corr}\big[C_\gamma(t), \mathrm{PLV}(t)\big] \gg 0. \tag{17}$$

Failure to observe any such correlation under well-controlled conditions would challenge the proposed link between Λ and Φ.

B.2 Gravimetric micro-perturbations

Biogravimetry within HLV predicts that coherent Φ-field dynamics generate small local gravitational perturbations of order $\Delta g \sim 10^{-10}\,\mathrm{m/s^2}$. During highly coherent neural states (e.g. deep meditation with synchronized gamma oscillations), atom-interferometric gravimeters placed near the brain should register synchronized micro-signals temporally aligned with peaks in $\mathcal{M}$.

B.3 Pharmacological modulation of Φ-coupling

Psychiatric medications that improve stability (e.g. lithium, lamotrigine) are expected to modify $g_{\Phi\Psi}$ and/or κ in Eq. (4). The framework predicts dose-dependent changes in both:

- biophotonic emission spectra and coherence times,
- macroscopic coherence measures (EEG/MEG).

If clinical stabilization occurs with no detectable change in any Φ-related observable, the interpretation of Λ as Φ would have to be reconsidered.

B.4 Persistence of coherence after intervention

System-independence requires that, after successful intervention, the system remains in a high-$\mathcal{M}$ state even after external support is removed. In our model this implies that $\Lambda(t)$ and the underlying Φ-configuration are reorganized into a new attractor state. Longitudinal studies should observe

$$\mathcal{M}_{\mathrm{post}} > \mathcal{M}_{\mathrm{baseline}} \tag{18}$$

even after tapering medication or other interventions. Persistent changes in Φ-linked observables would support the view that therapy has reconfigured the underlying field architecture rather than merely suppressing symptoms.

B.5 Falsification criteria

The unified framework is falsifiable if:

- no statistically significant relation can be established between Φ-sensitive observables (biophotons, gravimetric signals) and coherence measures derived from $\mathcal{M}$;
- interventions that change Γ, Δ, Θ in predictable ways produce no corresponding change in any Φ-linked quantity;
- purely classical electromagnetic models provide an equal or superior explanation of Λ-dynamics without invoking an HLV-type information field.

C Conclusion: From Metaphor to Mechanism

We have proposed a unified ($\mathcal{M}$–Φ) framework in which consciousness is understood as coherence. The Neuro-Coherence Function $\mathcal{M}$ specifies what consciousness *does*: it maintains stable information processing across scales through the interplay of adaptive gain, desynchronization, metabolic stability and spatiotemporal coherence. The HLV Φ-field specifies what consciousness *is*: quantum-informational field dynamics in biological tissue.

By identifying $\Lambda = \Phi$, we connect these perspectives into a single architecture:

$$\mathcal{U}_{\text{Consciousness}} = \mathcal{M}(\Gamma, \Delta, \Theta, \Phi). \tag{19}$$

This unification:

- transforms psychiatric disorders from purely chemical to coherence disorders,
- gives physics a direct biological realization of quantum informational fields,
- turns consciousness research into a domain with clear, testable mechanisms.

The next steps are clear: formal refinement of the coupled $(\mathcal{M}, \Phi)$ dynamics, computational modeling of brain-scale coherence trajectories, and experimental campaigns to test the predictions laid out in Sec. B. If validated, the framework could open a path toward genuine systemic cures for mental illness and a physics-based understanding of mind.

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Institutional Note

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Author Contributions

M. Krüger developed the Helix–Light–Vortex (HLV) framework, including all physical derivations, mathematical formulations, field-theoretic modelling, and the complete construction of the (M–Φ) coherence architecture. All core equations, dynamical laws, and theoretical interpretations originate from his foundational work.

D. Feeney contributed the systemic neuro-coherence operator formalism and the multi-scale modeling framework that links neural stability, metabolic regulation, and phase synchronization. His prior development of the Neuro-Coherence Function (M) provided the macroscopic structure into which the HLV field Φ is embedded. He also co-developed several cross-scale interpretations and clinical mappings in the present manuscript.

R. Duarte contributed conceptual and structural refinements to the coherence model. Specifically, he introduced the phase-constraint architecture formalized in Sec. 5, including the boundary-alignment logic, constraint manifolds, and coherence-propagation criteria. He also assisted in constructing the falsifiable test structure presented in Appendix A. These contributions sharpen the mathematical interpretation of coherence flow and stability conditions in the unified theory.

All scientific interpretations, mathematical structures, and final formulations were jointly reviewed and approved by all contributors.

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Declaration of Generative AI Assistance

During the preparation of this manuscript, the author used ChatGPT (OpenAI GPT-5) only for language polishing, grammar improvements, LaTeX formatting, and figure caption editing. All scientific ideas, theoretical concepts, derivations, analyses, and conclusions originate solely from the author. The author reviewed and edited all AI-assisted text and takes full responsibility for the content of this work.

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